APSC 2113 - Engineering Analysis I

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Homework 2 - Fall 2026

Homework Instructions

- The homework is due on **October 8, 2026, 11:59 PM.**
- Answer all questions. Simplify all answers completely.
- You may collaborate and discuss the homework with other students. However, each student must submit their own answers in their own submission.
- You may use the class notes, textbook, and course notes available on the course webpage to complete the homework. The use of AI tools is not permitted for this homework. You may, however, use Wolfram Alpha (or similar computational tools) to check or tune your understanding of concepts, not to generate your written solutions.
- All graphs must be drawn by hand and not generated using a computer or plotting software. The graphs are not being assessed for numerical precision but for whether you can demonstrate your understanding of the solution's behavior. You may use a calculator to compute specific (x, y) coordinate values if needed to guide your hand-drawn sketch.
- Write your name and student ID clearly at the top of the page.

Questions

Q1. Find the particular and homogeneous solution of $y' - 2y = e^{st}$ with initial state of $y(0) = 1$. **[1 pt]**

Q2. Use the integrating factor approach to solve $y' - 2y = e^{st}$ to demonstrate that both **Q1** and **Q2** approaches lead to the same solution. **[1 pt]**

Q3. For questions **Q3**, **Q4**, and **Q5**, apply the value of the parameter s to the solution obtained from **Q2** and provide responses to each sub-question.

$$s = 2.01$$

(a) Write down the solution for s = 2.01
(b) Draw a graph by hand approximately, and show what happens to y(t) with $s = 2.01$ for $0 \leq t \leq 2$

[1 pt]

Q4.

$$s = 2$$

(a) What happens to the solution at $s = 2$?
(b) Show any modifications necessary that you would make to the solution
(c) Graph the modified solution approximately by hand with $s = 2$ for $0 \leq t \leq 2$

[1.5 pt]

Q5.

$$s = 4$$

(a) Write down the solution for $s = 4$
(b) Draw a graph of the solution approximately by hand, with $s = 4$ for $0 \leq t \leq 2$

[1 pt]

Q6. Solve for initial state $y(0) = \frac{1}{2}$ using the partial fractions approach to integrate the relevant term

$$\frac{dy}{dt} = 4y - y^2$$

[1 pt]

Q7. Solve **Q6** using $y = \frac{1}{z}$ that transforms the original non-linear, first-order, homogeneous ODE into a linear first-order homogeneous ODE. Show that the approaches in **Q6** and **Q7** lead to the same solution. **[1 pt]**

Q8. Draw by hand the approximate graph of the solution to the non-linear, homogeneous ODE in **Q6**. Mark the steady states, initial states, and comment on the stability of the steady states. **[1 pt]**

Q9. Solve

$$\frac{dy}{dt} = 4y - y^3$$

using partial fractions to integrate the relevant term. **[1.5 pt]**

Q10. Answer the following questions about a second-order, linear, homogeneous ODE with constant coefficients: **[2 pt]**

$$\frac{d^2y}{dt^2} = -9y$$

(a) Sketch a spring-mass-damper system that is representative of the ODE.
(b) What is the spring stiffness constant per unit mass and the damping constant per unit mass?
(c) What is the natural frequency (ω_n) of this system.
(d) Solve $\frac{d^2y}{dt^2} = -9y$ and write the solution in terms of sine and cosine.
(e) What is the solution for the initial states y(0)=0 and y'(0)=1

Q11. Solve the following ODE and answer the questions related to the same **[2 pt]**

$$y'' - 5y' + 6y = 0$$

(a) Show all steps that lead to the general solution of the ODE.
(b) Apply initial conditions y(0)=1 and y'(0)=10. Write down the final solution
(c) Draw an approximate hand-drawn graph of the solution as $t \to \infty$, clearly marking the initial condition on it.
(d) What is the damping coefficient (per unit mass) of this system, and what is the damping ratio of this system?
(e) Explain in one or two sentences what effect the damping ratio has on the solution. Does it grow or decay unboundedly? Can you draw any conclusions from **Q11(c) and Q11(d)**

Q12. Solve the following ODE and answer the questions related to the same: **[1 pt]**

$$y'' - 6y' = 0$$

(a) Show all the steps that lead to the general solution of the ODE
(b) Apply initial conditions y(0)=0, y'(0)=1 and write the final solution.
(c) Sketch a mass-damper system that best represents the ODE

Q13. A 1kg mass is attached to a spring with k = 10 N/m and a damper with c = 2 Ns/m. It is displaced by 0.1 m and released from rest. **[2 pt]**

(a) Write down the governing ODE equation of motion
(b) Write down the initial states of the ODE
(c) Show all the steps that lead to the general solution of the ODE
(d) Apply the initial states and present the final solution
(e) What is the natural frequency (ω_n) of the system
(f) What is the damping ratio?

Q14. A 1 kg mass is attached to a spring with k = 9 N/m and a damper with c = 6 Ns/m. It is displaced by 0.1 m and released from rest. **[1.5 pt]**

(a) Write down the governing ODE for this problem
(b) Write down the initial states of the ODE
(c) Show that the damping ratio $\zeta = 1$
(d) Find the general solution of the ODE (show your steps)
(e) Apply initial states and show the final solution

Q15. A 2 kg mass rests at equilibrium on a spring with k = 58 N/m and a damper with c = 8 Ns/m. An initial velocity of 0.5 m/s is given to this oscillatory system. **[1.5 pt]**

(a) Show that the characteristic equation of the problem reduces to $s^2 + 4s + 29 = 0$.
(b) Find the natural frequency of this ODE
(c) What are the initial states of the ODE
(d) What is the solution of this ODE
(e) At what time does the first peak occur in this spring-mass-damper system once set in motion?

HW2

09/26/2026
KVB

①

Q① Solve $y' - 2y = e^{-5t}$ with initial state $y(0) = 1$

Q(a) Find Particular solution + Homogenous solution &

Solution Q1

$$\frac{dy}{dt} - 2y = e^{-5t}$$

Particular Solution (y_p):

$$y_p = Ae^{-5t}$$

$$\frac{dy_p}{dt} - 2y_p = e^{-5t}$$

$$\Rightarrow \frac{dAe^{-5t}}{dt} - 2Ae^{-5t} = e^{-5t}$$

$$\Rightarrow -5Ae^{-5t} - 2Ae^{-5t} = e^{-5t}$$

$$\Rightarrow A(-5-2) = 1$$

$$\Rightarrow A = \frac{1}{-5-2} \Rightarrow \boxed{y_p = \left(\frac{1}{-5-2}\right)e^{-5t}}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dt} = 2y_h \Rightarrow \frac{dy_h}{y_h} = 2dt$$

$$\Rightarrow \int \frac{dy_h}{y_h} = 2\int dt + C$$

$$\ln(y_h) = 2t + C$$

$$\Rightarrow \boxed{y_h = Ce^{2t}}$$

Complete solution: $y(t) = y_p + y_h$

$$\boxed{y(t) = \left(\frac{1}{-5-2}\right)e^{-5t} + Ce^{2t}}$$

Apply Initial State: $y(0) = 1$

$$\Rightarrow 1 = \left(\frac{1}{-5-2}\right)e^{-5(0)} + Ce^{2(0)}$$

$$\Rightarrow C = 1 - \left(\frac{1}{-5-2}\right) = \frac{-5-2-1}{-5-2} = \frac{-5-3}{-5-2}$$

$$y(t) = \left(\frac{1}{-5-2}\right)e^{-5t} + \left(\frac{-5-3}{-5-2}\right)e^{2t}$$

$$\boxed{y(t) = \frac{e^{-5t} + (-5-3)e^{2t}}{-5-2}} \rightarrow ①$$

Q② Use the Integrating factor approach to demonstrate that both approaches in Q① & Q② lead to the same solution (as they should!)

Solution Q2

Integrating factor: $e^{\int a(t)dt}$

General form of ODE: $\frac{dy}{dt} + a(t)y = q(t)$ — $a(t) = -2$, $q(t) = e^{-5t}$

$$\int a(t)dt = \int -2\,dt + C = -2t + C$$

$$e^{\int a(t)dt} = Ce^{-2t}$$

Multiplying by Ce^{-2t}

$$Ce^{-2t}\frac{dy}{dt} - 2Ce^{-2t}y = Ce^{-2t}e^{-5t}$$

$$d(e^{-2t}y) = e^{(-5-2)t}$$

$$\int d(e^{-2t}y) = \int e^{(-5-2)t} + C$$

$$e^{-2t}y = \frac{e^{(-5-2)t}}{-5-2} + C$$

$$\Rightarrow \boxed{y(t) = \frac{e^{-5t}}{-5-2} + Ce^{2t}}$$

Apply $y(0) = 1$: $y(0) = 1 = \frac{e^{-5(0)}}{-5-2} + Ce^{2(0)}$

$$\Rightarrow C = \frac{-5-3}{-5-2}$$

$$\Rightarrow \boxed{y(t) = \frac{e^{-5t}}{-5-2} + \frac{-5-3}{-5-2}e^{2t}} \rightarrow ②$$

① = ② QED ✓

For questions Q3, Q4 & Q5 apply the value for the parameter, s and provide responses to each sub-question.

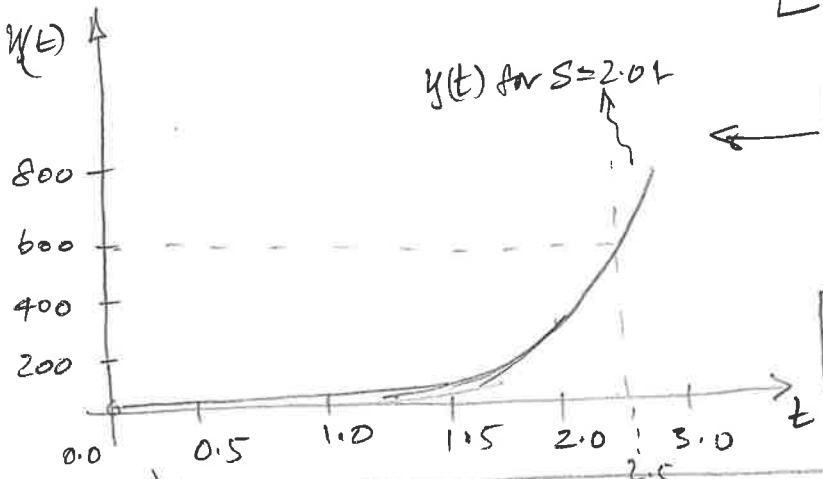
(Q3) $s = 2.01$ Solution: $y(t) = \frac{e^{st}}{s-2} + \frac{s-3}{s-2} e^{2t}$

$y(t) = \frac{e^{2.01t}}{2.01-2} + \left(\frac{2.01-3}{2.01-2}\right) e^{2t}$

$y(t) = \frac{e^{2.01t}}{0.01} - \frac{0.99}{0.01} e^{2t} = 100 e^{2.01t} - 99 e^{2t} = y(t)$

a) Write down the solution for $s = 2.01$

b) Draw a graph by hand, approximately, and show what happens to $y(t)$ with $s = 2.01$ for $0 \leq t \leq 2.5$



(Q4) $s = 2$ (a) What happens to the solution at $s = 2$? Ans: Resonance

(b) Show any modifications necessary you would make to the solution

Ans: Apply L'hôpital Rule $y(t) = \frac{e^{st} + (s-3)e^{2t}}{(s-2)}$

DUE TO RESONANCE

Numerator: $\frac{d}{ds}[e^{st} - (s-3)e^{2t}] = \frac{d}{ds}[e^{st} + se^{2t} + 3e^{2t}]$

$= te^{st} + e^{2t}$

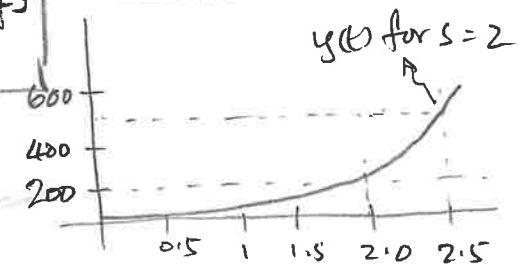
Denominator: $\frac{d}{ds}(s-2) = 1$

$\Rightarrow y(t) = \frac{\text{Numerator}}{\text{Denominator}} = \frac{te^{st} + e^{2t}}{1}$ $\Rightarrow y(t) = te^{st} + e^{2t}$

$\Rightarrow y(t) = te^{2t} + e^{2t}$

$\Rightarrow y(t) = (t+1)e^{2t}$

(c) Graph the modified solution, approximately, by hand with $s = 2$ for $0 \leq t \leq 2.5$



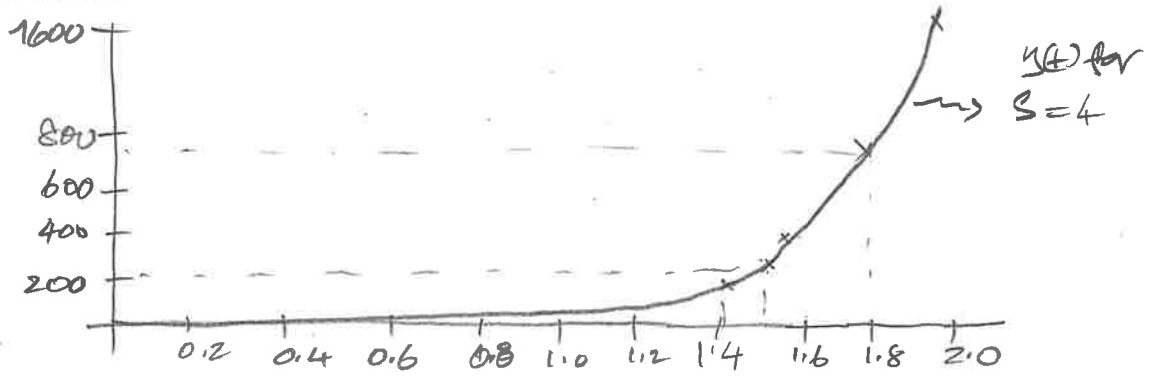
Q4 a) Write down the solution for $S = 4$ //

(3) KVB

Ans: $y(t) = \frac{e^{st}}{s-2} + \frac{s-3}{s-2} e^{2t} \Rightarrow y(t) = \frac{e^{4t}}{2} + \frac{1}{2} e^{2t}$

$\Rightarrow \boxed{y(t) = \frac{1}{2}(e^{4t} + e^{2t})}$

(b) Draw a graph of the solution approximately, by hand, with $S = 4$ & $0 \le t \le 2.0$ //



(86) Solve $\frac{dy}{dt} = 4y - y^2$ with initial state $y(0) = \frac{1}{2}$

Using Partial fractions approach

Ans: $\frac{dy}{dt} = 4y - y^2$

$\frac{dy}{dt} = y(4-y)$

$\Rightarrow \boxed{\frac{dy}{y(4-y)} = dt}$

Consider: $\frac{1}{y(4-y)} = \frac{A}{y} + \frac{B}{4-y}$

$\Rightarrow \frac{1}{y(4-y)} = \frac{A(4-y) + By}{y(4-y)}$

$\Rightarrow 4A - Ay + By = 1$

$\Rightarrow 4A = 1 \Rightarrow \boxed{A = \frac{1}{4}}$

& $-A + B = 0$

$\Rightarrow -\frac{1}{4} + B = 0$

$\Rightarrow \boxed{B = \frac{1}{4}}$

$\therefore \frac{dy}{y(4-y)} = \frac{1}{4y} + \frac{1}{4(4-y)}$

Integrating L.H.S term

$\int \frac{dy}{y(4-y)} = \int \frac{dy}{4y} + \int \frac{dy}{4(4-y)}$

$= \frac{1}{4}\ln(y) + \frac{1}{4}[-\ln(4-y)]$

$= \ln(y^{1/4}) - \ln(4-y)^{1/4}$

$\Rightarrow \int \frac{dy}{y(4-y)} = \ln\left\{\frac{y^{1/4}}{(4-y)^{1/2}}\right\}$

$\Rightarrow \int \frac{dy}{y(4-y)} = \ln\left(\frac{y}{4-y}\right)^{1/4} =$ R.H.S term

$\therefore \ln\left(\frac{y}{4-y}\right)^{1/4} = \int dt + C$

$\Rightarrow \ln\left(\frac{y}{4-y}\right)^{1/4} = t + C$

$\Rightarrow \left(\frac{y}{4-y}\right)^{1/4} = Ce^t$

$\Rightarrow \left(\frac{y}{4-y}\right) = Ce^{4t}$

$\Rightarrow (4-y)Ce^{4t} = y$

$\Rightarrow y = 4Ce^{4t} - yCe^{4t}$

$\Rightarrow y(1 + Ce^{4t}) = 4Ce^{4t}$

$\Rightarrow \boxed{y(t) = \frac{4Ce^{4t}}{1 + Ce^{4t}}}$

Apply $y(0) = 1/2$

$y(0) = 1/2 = \frac{4C}{1+C}$ ($e^{4(0)} \to 1$)

$\Rightarrow 8C = 1 + C \Rightarrow \boxed{C = 1/7}$

$\therefore y(t) = \frac{(4/7)e^{4t}}{1 + (1/7)e^{4t}}$

$\therefore \boxed{y(t) = \frac{4e^{4t}}{7 + e^{4t}}}$ → (1)

(Q7) Solve (Q6) using $y = \frac{1}{z}$ that transforms the original non-linear, first order, non-homogeneous ODE into a linear first-order, homogenous ODE. Show that both approaches in (Q6) & (Q7) lead to the same solution.

Ans: $\boxed{\frac{dy}{dt} = 4y - y^2}$

Let $z = \frac{1}{y}$

$\Rightarrow \frac{dz}{dt} = -\frac{1}{y^2}\frac{dy}{dt}$

$\Rightarrow \frac{dz}{dt} = -\frac{1}{y^2}(4y - y^2)$

$= -\frac{4y}{y^2} + \frac{y^2}{y^2}$

$= -\frac{4}{y} + 1$

$\Rightarrow \boxed{\frac{dz}{dt} = -4z + 1}$

Particular solution (z_p):

$z_p = A$ (say)

$\Rightarrow \frac{dz_p}{dt} = -4z_p + 1$

$\Rightarrow \frac{dA}{dt}^{0} = -4A + 1$

$\Rightarrow \boxed{A = 1/4} \Rightarrow \boxed{z_p = 1/4}$

Homogenous solution (z_h):

$\frac{dz_h}{dt} = -4z_h$

$\Rightarrow \frac{dz_h}{z_h} = -4\,dt$

Integrating

$\Rightarrow \int \frac{dz_h}{z_h} = \int -4\,dt + C$

$\Rightarrow \ln(z_h) = -4t + C$

$\Rightarrow \boxed{z_h = Ce^{-4t}}$

Complete solution for z: $z = z_p + z_h$

$\Rightarrow z = 1/4 + Ce^{-4t} = \frac{1 + 4Ce^{-4t}}{4}$

$\Rightarrow y = \frac{1}{z} = \frac{4}{1 + 4Ce^{-4t}}$

$\Rightarrow \boxed{y(t) = \frac{4e^{+4t}}{e^{4t} + 4C}}$

Apply $y(0) = 1/2$:

$\Rightarrow y(0) = 1/2 = \frac{4e^{4(0)}}{e^{4(0)} + 4C}$ ($e^{4(0)} \to 1$)

$\Rightarrow 1/2 = \frac{4}{1 + 4C} \Rightarrow 4C + 1 = 8$

$\Rightarrow \boxed{C = 7/4}$

$\therefore y(t) = \frac{4e^{4t}}{e^{4t} + 4(7/4)}$

$\therefore \boxed{y(t) = \frac{4e^{4t}}{e^{4t} + 7}} \rightarrow (2)$

$\therefore (1) = (2)$ ✓

(Q8) Draw/the graph of the solution to the non-linear, homogeneous ODE in (Q6) by hand → approximate and add mark the steady states, initial state and comment on the stability of the steady state.

Ans:

$$\boxed{y(t) = \frac{4e^{4t}}{7 + e^{4t}}}$$

$$\frac{dy}{dt} = 4y - y^2 \quad \Rightarrow \quad f(y) = 4y - y^2$$

(*) Set $f(y) = 0$ to identify steady states

$\therefore 4y - y^2 = 0 \Rightarrow y(4-y) = 0$

$\Rightarrow \boxed{y=0} \quad \boxed{y=4}$

Two steady states

(ii) Stability of $\boxed{y=0}$ & $\boxed{y=4}$:

$$\left.\frac{df}{dy}\right|_{y=0} = \left.\frac{d}{dy}(4y-y^2)\right|_{y=0} = \left.(4-2y)\right|_{y=0}$$

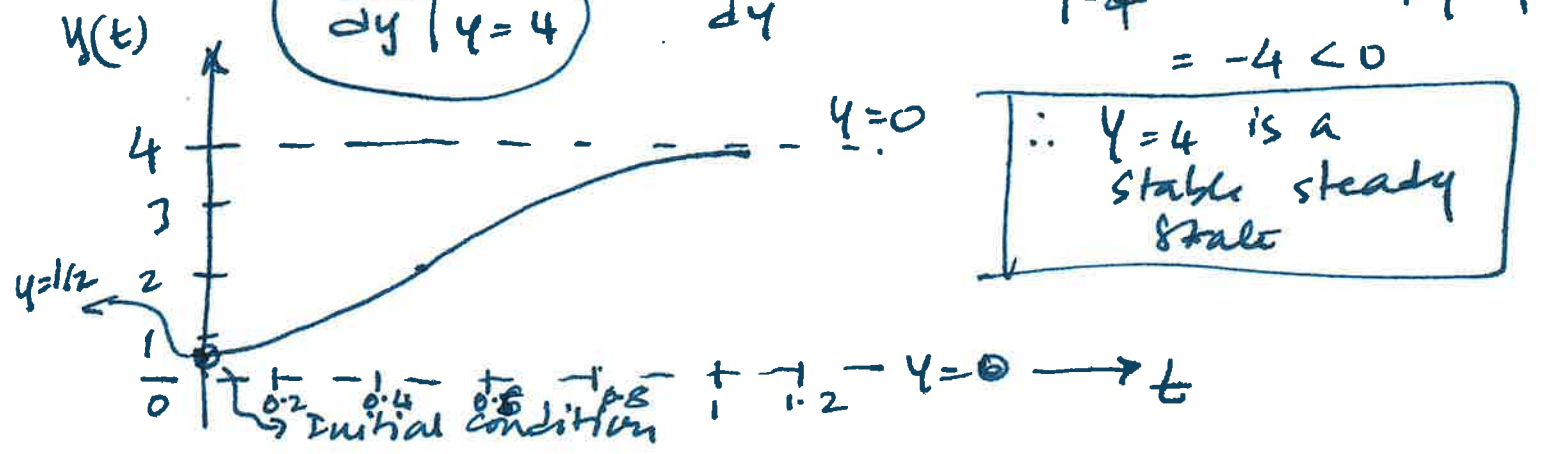
$$\Rightarrow \left.\frac{df}{dy}\right|_{y=0} = 4 - 0 = 4 > 0 \quad \therefore$$

$\boxed{y=0}$ is an unstable steady state

$$\left.\frac{df}{dy}\right|_{y=4} = \left.\frac{d}{dy}(4y-y^2)\right|_{y=4} = \left.(4-2y)\right|_{y=4}$$

$$= -4 < 0$$

$\therefore \boxed{y=4}$ is a stable steady state



(Q9) Solve $\frac{dy}{dt} = 4y - y^3$ using partial fractions for integrating the relevant term

Ans:

$$\frac{dy}{dt} = y(4-y^2)$$

$$\boxed{\frac{dy}{dt} = y(2+y)(2-y)}$$

$$\Rightarrow \frac{dy}{y(2+y)(2-y)} = dt \longrightarrow (1)$$

Consider the term: $\frac{1}{(y)(2+y)(2-y)}$

$$\frac{1}{(y)(2+y)(2-y)} = \frac{A}{y} + \frac{B}{2+y} + \frac{C}{2-y}$$

$$\frac{1}{(y)(2+y)(2-y)} = \frac{A(2+y)(2-y) + B(y)(2-y) + C(y)(2+y)}{A(y)(2+y)(2-y)}$$

$$\Rightarrow 1 = A(4-y^2) + B(2y-y^2) + C(2y+y^2)$$

$$\Rightarrow 1 = 4A - Ay^2 + 2By - By^2 + 2Cy + Cy^2$$

y^0: $4A = 1 \Rightarrow \boxed{A = 1/4}$

y^1: $2B + 2C = 0 \Rightarrow B + C = 0$

y^2: $-A - B + C = 0 \Rightarrow -1/4 - B + C = 0 \Rightarrow -B + C = 1/4$

Adding $B + C = 0$

$2C = 1/4 \Rightarrow \boxed{C = 1/8}$

$\Rightarrow \boxed{B = -1/8}$

$$\therefore \frac{1}{(y)(2+y)(2-y)} = \left(\frac{1}{4}\right)\left(\frac{1}{y}\right) - \left(\frac{1}{8}\right)\left(\frac{1}{2+y}\right) + \left(\frac{1}{8}\right)\left(\frac{1}{2-y}\right)$$

$$\int \frac{dy}{(y)(2+y)(2-y)} = \int\left(\frac{1}{4}\right)\left(\frac{dy}{y}\right) - \int\left(\frac{1}{8}\right)\left(\frac{dy}{2+y}\right) + \int\left(\frac{1}{8}\right)\left(\frac{dy}{2-y}\right)$$

$$\int \frac{dy}{(y)(2+y)(2-y)} = \left(\frac{1}{4}\right)\ln(y) - \left(\frac{1}{8}\right)\ln(2+y) + \left(\frac{1}{8}\right)\left[-\ln(2-y)\right]$$

$$= \ln(y^{1/4}) + \ln(2+y)^{-1/8} + \ln(2-y)^{-1/8}$$

$$= \ln\left[(y^{1/4})(2+y)^{-1/8}(2-y)^{-1/8}\right]$$

$$= \ln\left[\frac{y^{1/4}}{(4-y^2)^{1/8}}\right] \longrightarrow \text{L.H.S of eq (1)}$$

$$\ln\left[\frac{y^{1/4}}{(4-y^2)^{1/8}}\right] = \int dt + c$$

$$= t + c$$

$$\Rightarrow \frac{y^{1/4}}{(4-y^2)^{1/8}} = ce^t \Rightarrow \frac{(y^2)^{1/8}}{(4-y^2)^{1/8}} = ce^t$$

$$\Rightarrow \left(\frac{y^2}{4-y^2}\right)^{1/8} = ce^t$$

$$\Rightarrow \left(\frac{y^2}{4-y^2}\right) = ce^{8t}$$

$$\Rightarrow y^2 = ce^{8t}(4-y^2) = 4ce^{8t} - y^2ce^{8t}$$

$$\Rightarrow y^2(1+ce^{8t}) = 4ce^{8t}$$

$$\Rightarrow y^2 = \frac{4ce^{8t}}{1+ce^{8t}} \Rightarrow y = \pm\frac{(4ce^{8t})^{1/2}}{(1+ce^{8t})^{1/2}}$$

$$y = \pm\frac{(4)^{1/2}}{\left(\frac{1+ce^{8t}}{ce^{8t}}\right)^{1/2}} = \pm\frac{2}{\left(\frac{1}{c}e^{-8t}+1\right)^{1/2}}$$

$$\boxed{y(t) = \pm\frac{2}{\sqrt{1+Ke^{-8t}}}}$$

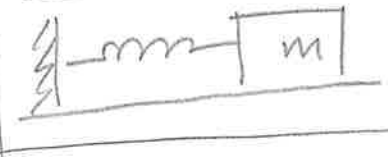
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(Q10) Answer the following questions about a second-order, linear, homogenous ODE with constant coefficients:

(a) Sketch a spring-mass-damper system that is representative of the ODE

$$\frac{d^2y}{dt^2} = -9y$$

Ans: 10a



(c) What is the natural frequency (ω_n) of this system

Ans: 10c

General form of the 2nd order, linear, homogenous ODE

$$y'' + Ay' + By = 0$$

$$y'' + \frac{k}{m}y = 0 \Rightarrow y'' = -9y$$

$$\sqrt{\frac{k}{m}} = \omega_n \Rightarrow \omega_n = \sqrt{9}$$

$$\omega_n = 3$$

(b) What is the spring stiffness constant per unit mass and the damping constant per unit mass?

Ans: 10b

$\frac{k}{m} = 9$ $\quad$ $\frac{c}{m} = 0$

(d) Solve $\frac{d^2y}{dt^2} = -9y$ and write the solution in terms of sine and cosine

Solution:

(1) Characteristic equation $\quad s^2 + 9 = 0$

(2) Roots: $s = \pm\sqrt{-9} \Rightarrow$ $s_1 = +3i$, $s_2 = -3i$

(d) General form of the solution

$$y = c_1 e^{s_1 t} + c_2 e^{s_2 t}$$

$$\therefore y = c_1 e^{i3t} + c_2 e^{-i3t}$$

$$\Rightarrow y = c_1 [\cos 3t + i \sin 3t] + c_2 [\cos 3t - i \sin 3t]$$

$$\Rightarrow y = (c_1 + c_2) \cos 3t + i (c_1 - c_2) \sin 3t$$

$$\Rightarrow \boxed{y(t) = c_1 \cos 3t + c_2 \sin 3t}$$

(e) Write the solution for the initial states.

$y(0) = 0$, $y'(0) = 1$

$$y(0) = c_1 \cos 3(0) + c_2 \sin 3(0) = 0$$

($\cos 3(0) \to 1$, $\sin 3(0) \to 0$)

$$\Rightarrow \boxed{c_1 = 0}$$

$$y'(t) = -3c_1 \sin 3t + 3c_2 \cos 3t$$

($-3c_1 \sin 3t \to 0$)

$$y'(0) = 3c_2 \cos(0) = 1 \Rightarrow 3c_2 = 1$$

($\cos(0) \to 1$)

$$\Rightarrow \boxed{c_2 = 1/3}$$

$$\therefore \boxed{y(t) = \frac{1}{3} \sin 3t}$$

(Q11) Solve the following ODE and answer the questions related to the same

$$y'' - 5y' + 6y = 0$$

(a) Show all steps that lead to the general solution of the ODE

Ans: (i) Characteristic equation: $s^2 - 5s + 6 = 0$

$\Rightarrow (s-2)(s-3) = 0$

$\boxed{s_1 = 2}$; $\boxed{s_2 = 3}$

(ii) General solution: $\boxed{C_1 e^{2t} + C_2 e^{3t} = y(t)}$

(b) Apply initial conditions $y(0) = 1$ & $y'(0) = 10$ and write down the final solution

Ans: $\boxed{y(0) = 1} \Rightarrow y(0) = C_1 e^{2(0)} + C_2 e^{3(0)} = 1$

$\Rightarrow \boxed{C_1 + C_2 = 1}$

$\boxed{y'(0) = 10} \Rightarrow y'(t) = 2C_1 e^{2t} + 3C_2 e^{3t}$

$\Rightarrow y'(0) = 2C_1 e^{2(0)} + 3C_2 e^{3(0)} = 10$

$= \boxed{2C_1 + 3C_2 = 10}$

$2C_1 + 3C_2 = 10$

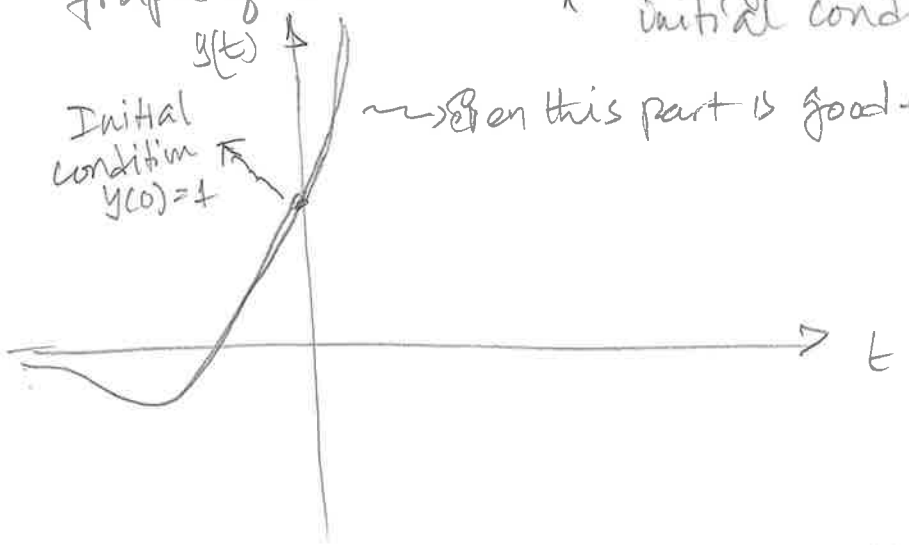
$-2C_1 + 2C_2 = 2 \quad \times 2$

$\boxed{C_2 = 8}$ & $\boxed{C_1 = -7}$

$\therefore \boxed{y(t) = -7e^{3t} + 8e^{3t}}$

QED

Q11 (d) Draw an approximate hand drawn graph of the solution as $t \to \infty$, clearly marking the initial condition on it



→ Only on this part is good enough

(e) What is the damping coefficient of this system (per unit mass) and what is the damping ratio of this system

Ans: $\frac{c}{m} = -5$

$$\boxed{\zeta = \frac{c}{2\sqrt{mk}}} \rightarrow \zeta = \frac{c/m}{\frac{2}{m}\sqrt{mk\frac{m}{m}}}$$

$$= \frac{-5}{\frac{2}{m}\,m\sqrt{\frac{k}{m}}}$$

$$= \frac{-5}{2\sqrt{6}} < 0$$

(f) Explain in one or two sentences, what effect does the damping ratio have on the solution? Does it grow or decay unboundedly? Can you draw any conclusions from Q11 d & e?

Ans: KEYWORDS: ① Unstable
② Solution grows without bound

(Q12) Solve the following ODE and answer the questions related to the same: $y'' - 6y' = 0$

(a) Show all the steps that lead to the general solution of the ODE

Ans: (i) Characteristic equation: $s^2 - 6s = 0$

$\Rightarrow s(s-6) = 0$

$s_1 = 0$ $\quad$ $s_2 = 6$

(ii) General Solution: $y = C_1 e^{0t} + C_2 e^{6t}$

$\Rightarrow \boxed{y(t) = C_1 + C_2 e^{6t}}$

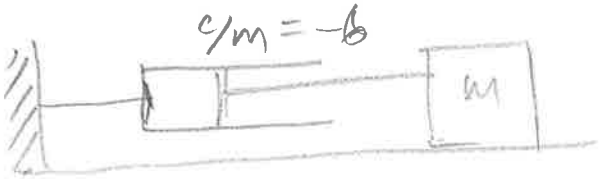
(b) Apply initial conditions $y(0) = 0$, $y'(0) = 1$ and write the final solution:

Ans: $y(0) = C_1 + C_2 e^{6(0)} = 0 \Rightarrow C_1 + C_2 = 0$

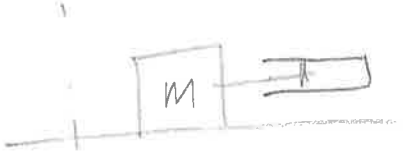
$y'(0) = 6C_2 e^{6(0)} = 1 \Rightarrow \boxed{C_2 = \frac{1}{6}} \Rightarrow \boxed{C_1 = -\frac{1}{6}}$

$\therefore y(t) = -\frac{1}{6} + \frac{1}{6} e^{6t} \Rightarrow \boxed{y(t) = \frac{1}{6}(e^{6t} - 1)}$

(c) Sketch a mass damper system that best represents the ODE



OR



(Q13) A 1Kg mass is attached to a spring with $k = 10\ N/m$ and a damper with $C = 2\ Ns/m$. It is displaced 0.1 m and released from rest.

a) Write down the governing ODE equation of motion:

Ans: $y'' + 2y' + 10y = 0$

b) Write down the initial states of the ODE

Ans: $y(0) = 0.1\ m$

$y'(0) = 0$

c) Show all the steps that lead to the general solution of the ODE

Ans: (I) Characteristic equation: $s^2 + 2s + 10 = 0$

$$s = \frac{-2 \pm \sqrt{2^2 - 4(1)(10)}}{2} = \frac{-2 \pm \sqrt{4-40}}{2}$$

$$\Rightarrow s = \frac{-2 \pm i\sqrt{36}}{2} = -1 \pm 3i$$

$$\Rightarrow \boxed{s_1 = -1 + 3i} \text{ & } \boxed{s_2 = -1 - 3i}$$

(II) General solution: $y(t) = c_1 e^{s_1 t} + c_2 e^{s_2 t}$

$$y(t) = c_1 e^{(-1+3i)t} + c_2 e^{(-1-3i)t}$$

$$\Rightarrow y(t) = e^{-t}(c_1 e^{i3t} + c_2 e^{-i3t})$$

$$\Rightarrow \boxed{y(t) = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)}$$

Q13
(d) Apply the initial states and present the final solution:

$\boxed{y(0) = 0.1}$

$\Rightarrow y(0) = e^{-(0)}(C_1 \cos(0) + C_2 \sin(0)) = 0.1$

$\Rightarrow \boxed{C_1 = 0.1}$

$\boxed{y'(0) = 0}$

$\Rightarrow y'(t) = e^{-t}(-C_1 3\sin 3t + C_2 3\cos 3t) - e^{-t}(C_1 \cos 3t + C_2 \sin 3t)$

$\Rightarrow y'(0) = e^{0}(-0.1 \times 3 \sin(0) + C_2 \times 3 \cos(0)) - e^{0}(C_1 \cos(0) + C_2 \sin(0))$

$0 = 3C_2 - 0.1 \Rightarrow \boxed{C_2 = \frac{1}{30}}$

$\therefore \boxed{y(t) = e^{-t}\left[(0.1)\cos 3t + \left(\frac{1}{30}\right)\sin 3t\right]}$

(e) What is the natural frequency (ω_n) of the system?

Ans: $\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{10}{1}} = 3.162 \text{ rad/s}$

(f) What is the damping ratio?

Ans: $\zeta = \frac{C}{2\sqrt{mK}} = \frac{2}{2\sqrt{10}} \approx 0.316$

$0 < \zeta < 1 \therefore$ Underdamped

(Q 14) A 1kg mass is attached to a spring with $k = 9$ N/m and a damper with $C = 6$ Ns/m. It is displaced 0.1m and released from rest.

(1ST KVB)

(a) Write down the governing ODE for this problem

Ans: $y'' + 6y' + 9y = 0$

(b) Write down the initial states of the ODE

Ans: $y(0) = 0.1$ m

$y'(0) = 0$

(c) Show that the damping ratio, $\zeta = 1$:

$$\zeta = \frac{c}{2\sqrt{mk}} = \frac{6}{2\sqrt{(1)9}} = \frac{6}{2(3)} = 1 \quad \checkmark$$

$\therefore \zeta = 1 \Rightarrow$ Critically damped

(d) Find the general solution of the ODE: (show your steps)

(I) Characteristic equation:

$$s^2 + 6s + 9 = 0 \Rightarrow (s+3)^2 = 0$$

$$\Rightarrow s_1 = -3, \; s_2 = -3 \} \text{ repeated roots}$$

(II) General solution: we have repeated roots

$$\therefore \boxed{y(t) = (C_1 + C_2 t)e^{-3t}}$$

Q14

(e) Apply initial states and show the final solution:

$$y(0) = 0.1 = [C_1 + C_2(0)]e^{-3(0)}$$

$$\Rightarrow \boxed{C_1 = 0.1}$$

$$y'(t) = C_2 e^{-3t} - 3(C_1 + C_2 t)e^{-3t}$$

$$y'(0) = C_2 e^{0} - 3(C_1 + C_2(0))e^{0}$$

$$\Rightarrow C_2 - 3C_1 = 0$$

$$\Rightarrow C_2 = 3(0.1) \Rightarrow \boxed{C_2 = 0.3}$$

$$\therefore y(t) = (0.1 + 0.3t)e^{-3t}$$

$$\Rightarrow \boxed{y(t) = 0.1(1 + 3t)e^{-3t}}$$

Q15

(17)

KVB

A 2kg mass rests at equilibrium on a spring with $k = 58$ N/m and a damper with $C = 8$ N s/m. An initial velocity of 0.5 m/s is given to this oscillating system.

(a) Show that characteristic equation of the problem reduces to $s^2 + 4s + 29 = 0$

Ans: ODE: $2y'' + 8y' + 58y = 0$

$\Rightarrow y'' + 4y' + 29y = 0$

$\therefore$ characteristic equation: $\boxed{s^2 + 4s + 29 = 0}$ ✓

(b) Find the natural frequency of this ODE

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{58}{2}} = \sqrt{29} = 5.385 \text{ rad/s}$$

(c) What are the initial states of the ODE?

$y(0) = 0$

$y'(0) = 0.5$ m/s

(d) What is the solution of the ODE?

$s^2 + 4s + 29 = 0 \Rightarrow \boxed{s_1 = -2 + 5i}$, $\boxed{s_2 = -2 - 5i}$

$\therefore$ General solution: $y(t) = C_1 e^{(-2+5i)t} + C_2 e^{(-2-5i)t}$

$$y(t) = C_1 e^{(-2+5i)t} + C_2 e^{(-2-5i)t}$$

$$= e^{-2t}\left[C_1 e^{5it} + C_2 e^{-5it}\right]$$

$$= e^{-2t}\left[C_1 \cos 5t + C_2 \sin 5t\right]$$

Applying ICs: $y(0) = 0$

$$\Rightarrow y(0) = (e^{0})\left[C_1 \cos(0) + C_2 \sin(0)\right] = 0$$

($e^0 = 1$, $\cos(0) = 1$, $\sin(0) = 0$)

$$\Rightarrow \boxed{C_1 = 0}$$

$$y'(t) = e^{-2t}\left(-C_1 5 \sin 5t + C_2 5 \cos 5t\right)$$

$$y'(0) = 0.5 = e^{-0}\left(-C_1 \sin(0) + C_2 5 \cos(0)\right)$$

($\sin(0) = 0$, $\cos(0) = 1$)

$$\Rightarrow 5C_2 = 0.5 \Rightarrow \boxed{C_2 = 0.1}$$

$$\therefore y(t) = e^{-2t}(0.1)\sin 5t$$

$$\boxed{y(t) = 0.1 e^{-2t} \sin 5t}$$

(e) At what time does the first peak occur in this spring-mass-damper system once set on motion?

(#) When Peak occurs, slope = 0

$$\therefore y'(t) = (0.1)\left[e^{-2t}\cos 5t \cdot -2 e^{-2t}\sin 5t\right] = 0$$

$$= e^{-2t}\left[0.5\cos 5t - 0.2 \sin 5t\right) = 0$$

$$\Rightarrow \tan 5t = \frac{0.5}{0.2} = 2.5 \Rightarrow \therefore t = \frac{1}{5}\tan^{-1} 2.5$$

$$\Rightarrow \boxed{t \tilde{=} 0.238 \text{ seconds}}$$